Scalable Image-on-Scalar Regression for High-Dimensional Neuroimaging Data

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April 1, 2026

Keywords: image-on-scalar regression, spatially varying coefficients, scalable computation, neuroimaging, functional MRI

Abstract: Image-on-scalar regression is a statistical method for modeling the association between scalar covariates and an image outcome, which takes on different values at different spatial locations. Although many methods exist for this class of problems, the time and memory required to apply these methods in practice scale poorly with the sample size and the number of voxels in the image, both of which are typically very large for questions of interest. We propose a new method, the Scalable Image-on-Scalar Regression Algorithm (SIRA), for solving image-on-scalar regression problems with a specific focus on maintaining computational scalability in both the sample size and the image size by enforcing homogeneity and sparsity in the spatially varying coefficient solution to this problem. We compare the performance of SIRA to other existing image-on-scalar regression methods on simulated data and demonstrate superior performance of SIRA in terms of various metrics. We also provide an application to Functional Magnetic Resonance Imaging data from UK Biobank with 38,639 subjects and 121,865 voxels, which is beyond the scale that most other image-on-scalar regression methods can reliably handle.

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1 Introduction

Modern neuroimaging studies often aim to discover associations between brain activity and subject-level variables such as age, gender, or cognitive traits. In this framework, the entire image of brain intensity values is modeled as the outcome variable, and scalar covariates serve as predictors. A traditional approach for this class of problems, known as image-on-scalar regression, is mass univariate analysis (MUA) (Groppe et al., 2011), in which a regression model is fit independently on every voxel, and a false discovery rate or family-wise error rate adjustment is applied to control the number of false positives. This analysis identifies a set of voxels for which individual covariates are significantly associated with the corresponding brain intensity outcomes. While MUA has been widely used in neuroimaging research, it comes with the considerable drawback of ignoring valuable information in the form of the spatial structure of the brain voxels. Alternative methods that account for the adjacency of the voxels do exist and can perform well in certain contexts, but struggle with various computational challenges when they are applied to large-scale heterogeneous neuroimaging data.

One of the earliest modifications to MUA came from the Statistical Parametric Mapping (SPM) software (Ashburner, 2012). For the initial method developed in the 1990s, SPM first performs spatial normalization and smoothing via convolution with a Gaussian kernel, allowing for spatial information to be utilized. In the 2000s, SPM began incorporating more advanced statistical methods such as random field theory and Bayesian modeling for more flexible analysis (Friston et al., 2002). However, these voxel-wise methods generally have sub-optimal power (Chumbley & Friston, 2009), which is hindered even further by the use of Gaussian smoothing to blur important features of the image (Loh & Lindquist, 2010). A newer approach targeted at neuroimaging data with jump discontinuities is the Spatially Varying Coefficient Model (SVCM) (Zhu et al., 2014). SVCM capitalizes on the biological nature of neuroimaging data containing multiple piecewise smooth regions with sharp edges by proposing a jumping surface model.

Another more recent approach to the image-on-scalar regression problem is the bivariate spline functions over triangulation (BST) method (Li et al., 2021). This method extends the SVCM model into a partially linear spatially varying coefficient model (PLSVCM), improving estimation accuracy and enabling identification of components with spatially varying or constant values. A related spline-based approach (Yu et al., 2021) further develops this framework by establishing asymptotic properties of the coefficient estimators and constructing simultaneous confidence corridors, providing a rigorous inferential foundation for image-on-scalar regression. However, these methods are primarily applicable to two-dimensional imaging data. A more advanced approach suitable for three-dimensional imaging data is the image response regression via deep neural networks (IRRNN) method (Zhang et al., 2023), which was designed to have more flexibility in capturing complex spatial patterns in datasets consisting of a small number of

high-resolution images. In the Bayesian paradigm, a hierarchical Gaussian process model with a spatial global-local spike-and-slab prior has been proposed that simultaneously performs variable selection at both the covariate level and the individual voxel level, offering a fully model-based approach to inference and selection (Zeng et al., 2024). More recently, a scalable Bayesian image-on-scalar regression model has been proposed that employs stochastic gradient Langevin dynamics combined with memory-mapping techniques to accommodate large sample sizes, while also handling subject-specific brain masks through an imputation-based approach (Xu et al., 2026).

All of these aforementioned new methods have specific advantages or drawbacks in certain settings, but share one fundamental weakness that has become increasingly relevant over time. With the exception of MUA, the most rudimentary and least powerful of these methods, none of these methods can reliably handle working with large-scale datasets, a very desirable and arguably necessary property for any image-on-scalar regression method in the present age. When attempting to apply these methods on 180 GB of real functional magnetic resonance imaging (fMRI) data from UK Biobank (UKB) (Sudlow et al., 2015), none of them were able to successfully read in the fMRI dataset and produce a solution within 12 hours of runtime. While Xu et al. (2026) does address computational scalability, their method requires a high-performance computing cluster and 20.5 hours of runtime. This high dimensionality of imaging outcomes, coupled with growing sample sizes, necessitates the development of scalable algorithms that can efficiently handle massive data while maintaining accuracy and interpretability.

The recent rapid growth in neuroimaging datasets, largely driven by initiatives such as UKB, the Adolescent Brain Cognitive Development (ABCD) study (Casey et al., 2018), and the Human Connectome Project (Van Essen et al., 2012), poses a great challenge for existing image-on-scalar regression methods. Data from these sources can include hundreds of thousands of subjects, each with images comprising hundreds of thousands of voxels, resulting in ultra-high dimensional data. This unprecedented scale makes computational efficiency and scalability critical concerns. To address the shared computational limitations of the previous methods, we propose a novel algorithm, the Scalable Image-on-Scalar Regression Algorithm (SIRA), to solve image-on-scalar regression problems in settings with a large sample size, image size, or both, within a computationally feasible time frame. In the remainder of the manuscript, we introduce the underlying model and loss function for SIRA, describe the parameter estimation procedures, compare the performance of SIRA to other image-on-scalar regression methods using simulated data, and analyze the results of SIRA when applied to real UKB fMRI data.

2 Methods

2.1 Model

Suppose the data consist of observations from n subjects. Let $i \in \{1, \ldots, n\}$ index the subject. We assume the imaging measurements across all the subjects are registered to the standard compact space $\mathcal{B} \subset \mathbb{R}^d$. The imaging data are collected at V voxels, denoted as $\mathcal{S} = \{s_1, \ldots, s_V\} \subset \mathcal{B}$. Let $Y_i(s_v)$ denote the observed image intensity at voxel s_v for subject i. We also collect a set of $p = p_1 + p_2$ covariates. Let $j \in \{1, \ldots, p_1\}$ index the covariates of interest, and $k \in \{1, \ldots, p_2\}$ index the covariates of no interest. Let X_{ij} and Z_{ik} denote the covariate of interest j and covariate of no interest k for subject i, respectively. We consider the following image-on-scalar regression model:

$$Y_i(s_v) = \sum_{j=1}^{p_1} \beta_j(s_v) X_{ij} + \sum_{k=1}^{p_2} \gamma_k(s_v) Z_{ik} + \eta_i(s_v) + \epsilon_i(s_v), \tag{1}$$

with $\mathrm{E}\{\epsilon_i(s_v)\} = 0$ and $\mathrm{Var}\{\epsilon_i(s_v)\} = \sigma^2(s_v)$. The spatially varying coefficient $\beta_j(s_v)$ represents the spatial effects of the jth covariate of interest on the imaging outcome at s_v, and $\gamma_k(s_v)$ represents the spatial effects of the kth covariate of no interest on the imaging outcome at s_v. We make the following assumptions on the spatially-varying function $\beta_j(s_v)$.

1. (Sparsity) $\beta_j(s_v)$ is sparse; i.e. for each j, let $\mathcal{R}_{0,j}$ be the set of voxels s_v for which $\beta_j(s_v) = 0$ and let $\mathcal{R}_{1,j}$ be the set of voxels s_v for which $\beta_j(s_v) \neq 0$. For each j, the cardinality of $\mathcal{R}_{0,j}$ is much larger than that of $\mathcal{R}_{1,j}$.

2. (Homogeneity) $\beta_j(s_v)$ is a piecewise constant function and only takes a set of M_j distinct values, denoted as $\{\beta_{j,m}\}_{m=1}^{M_j}$. That is, there exists a partition of $\mathcal{B}$, denoted as $\bigcup_{m=1}^{M_j} \mathcal{S}_{j,m} = \mathcal{B}$ and $\mathcal{S}_{j,m'} \cap \mathcal{S}_{j,m} = \emptyset$ if $m \neq m'$, such that

$$\beta_j(s_v) = \sum_{m=1}^{M_j} \beta_{j,m} I(s_v \in \mathcal{S}_{j,m}).$$

Essentially, $\beta_j(s_v)$ is assumed to be a sparse and homogeneous function for all j, meaning it is identically zero at a majority of voxel locations, and otherwise forms a small number of regions of constant value. These regions $\mathcal{S}_{j,m}$ are learned during model fitting and not an input parameter of the model. Regions for different covariates j, j' exist independently of each other, i.e., if voxels s_v and $s_{v'}$ exist in the same region for covariate j, they are not necessarily in the same region for covariate j'. These assumptions

allow for both ease of computation and interpretability in our resulting estimate for β. We do not make any assumptions on the form of $\gamma_k(s_v)$.

The term $\eta_i(s_v)$ represents the spatial individual effects that are not explained by the covariates. We model η_i as a Gaussian process, $\mathcal{GP}(0, \kappa)$. The covariance function is assumed to be a modified squared exponential kernel, given by $\kappa_{MSE}(s_v, s_{v'}) = \exp\{-a(||s_v||^2 + ||s_{v'}||^2) - b||s_v - s_{v'}||^2\}$, for $a > 0, b > 0$. $||\cdot||$ represents the L_2-norm. The larger the value of a, the more the Gaussian process is concentrated around the center. The smaller the value of b, the smoother the Gaussian process is. As η_i is assumed to be spatially correlated, ϵ_i is assumed to be independent across all locations s_v. We also assume a constant value of $\sigma^2(s_v)$ regardless of location, i.e. $\sigma^2(s_v) = \sigma^2$ for all $s_v \in \mathcal{S}$.

By Mercer's Theorem (Rasmussen & Williams, 2006), the covariance kernel can be decomposed as $\kappa_{MSE}(s_v, s_{v'}) = \sum_{l=1}^{\infty} \lambda_l \psi_l(s_v)\psi_l(s_{v'})$, where the λ_l are eigenvalues in descending order and $\psi_l(v)$ are orthonormal eigenfunctions satisfying the property $\int_{\mathbb{R}^d} \psi_l(s_v)\psi_l(s_{v'})ds_v = 1$ when $s_v = s_{v'}$ and 0 otherwise. This allows us to construct our Gaussian process as $\eta_i(s_v) = \sum_{l=1}^{\infty} \theta_{il}\sqrt{\lambda_l}\psi_l(s_v)$, where $\theta_{il} \sim N(0, \tau^2)$. We approximate the Gaussian process by truncating the eigen-decomposition to L basis functions, i.e., $\eta_i(s_v) \approx \sum_{l=1}^{L} \theta_{il}\sqrt{\lambda_l}\psi_l(s_v)$. L is determined by the highest degree, h, of Hermite polynomials used in the eigen-decomposition, via the formula $L = \binom{h+d}{d}$.

For ease of notation, let $\psi_l^*(s_v) = \sqrt{\lambda_l}\psi_l(s_v)$. Thus, our image-on-scalar regression model can be rewritten as:

$$Y_i(s_v) = \sum_{j=1}^{p_1} \beta_j(s_v)X_{ij} + \sum_{k=1}^{p_2} \gamma_k(s_v)Z_{ik} + \sum_{l=1}^{L} \theta_{il}\psi_l^*(s_v) + \epsilon_i(s_v), \tag{2}$$

where the parameters we seek to estimate are the $p_1 \times V$ matrix β, the $p_2 \times V$ matrix γ, and the $n \times L$ matrix θ.

2.2 Loss Function

To estimate our set of parameters, we attempt to minimize a loss function that incorporates both prediction error and our assumptions on β. Let $\tilde{Y} = Y - Z\hat{\gamma} - \hat{\eta} = Y - Z\hat{\gamma} - \hat{\theta}\psi^*$. We propose the following loss function:

$$L(\hat{\beta}, \hat{\gamma}, \hat{\theta}) = \frac{1}{n} \sum_{i=1}^{n} \sum_{s_v \in S} \left[\tilde{Y}_i(s_v) - \sum_{j=1}^{p_1} X_{ij} \hat{\beta}_j(s_v) \right]^2 + \lambda \sum_{j=1}^{p_1} \sum_{(s_a, s_b) \in E} |\hat{\beta}_j(s_a) - \hat{\beta}_j(s_b)| + \mu \sum_{j=1}^{p_1} \sum_{s_v \in \mathcal{S}} |\hat{\beta}_j(s_v)|, \tag{3}$$

where E represents the set of edges in the image, i.e., all pairs of neighboring voxels. For the purposes of a 3-dimensional image, we treat every interior voxel as having 6 neighboring voxels (left, right, above, below, front, back), while boundary voxels will have fewer than 6 neighbors. λ and μ are tuning parameters that control the strength of the penalties in the second and third terms of the loss function.

The first term is a standard mean squared error, where we have absorbed multiple terms together into the variable $\tilde{Y}$. The second term places a penalty on the absolute value of the difference in the β values of neighboring voxels, over all possible neighbor pairs. This promotes homogeneous regions of β values, as the absolute value shrinks the difference in β values of neighboring voxels to exactly zero. This concept is similar to the fused lasso penalty used in standard linear regression. The third term places a L_1 penalty on the β values, which promotes sparsity in our β estimate, as values close to zero will be shrunk to exactly zero. Together, these two additional terms in our loss function help promote an estimate for β that is consistent with the original assumptions we placed on β.

2.3 Parameter Updating Algorithm

We propose a block coordinate descent-type algorithm to estimate our parameters β, γ, and θ. Given initial values $\hat{\beta}, \hat{\gamma}$, and $\hat{\theta}$, we fix the values of two of them and optimize the value of the third, and repeat this process, cycling through which variable is being optimized. We continue this process until a convergence criterion has been met. Because our loss function is the sum of three convex terms, the overall loss function is also convex, which eliminates the local minima problem because every local minimum is also a global minimum. The outline of this procedure is given in Algorithm 1.

Details on the region finding algorithm and least squares updates are given in subsection 2.4. For the convergence criterion, we set $\delta = 0.01$. We also impose a hard cap of $T_{\max} = 50$ iterations to guarantee termination; in practice, convergence is reached well before this limit.

Algorithm 1: Scalable Image-on-Scalar Regression Algorithm (SIRA)

Input: Data Y, X, Z; tuning parameters λ, μ; convergence threshold δ; maximum iterations $T_{\max}$

Output: Estimates $\hat{\beta}$, $\hat{\gamma}$, $\hat{\theta}$

Initialize $\hat{\beta}$, $\hat{\gamma}$, $\hat{\theta}$ (see section 2.5), and iteration counter $t \leftarrow 0$;

while Convergence criterion not met: $t < T_{\max}$ and $\|\hat{\alpha}^{(t)} - \hat{\alpha}^{(t-1)}\|_F \geq \delta$, where $\hat{\alpha} = (\hat{\beta}^\top, \hat{\gamma}^\top)^\top$ **do**

  for $j = 1, \ldots, p_1$ **do**

    Fix $\hat{\gamma}$ and $\hat{\theta}$;

    Update $\hat{\beta}_j$ via the region finding algorithm (Algorithm 2);

  end

  Fix $\hat{\beta}$ and $\hat{\theta}$; update $\hat{\gamma}$ via closed-form least squares;

  Fix $\hat{\beta}$ and $\hat{\gamma}$; update $\hat{\theta}$ via closed-form least squares;

  Set $t \leftarrow t + 1$;

end

return $\hat{\beta}$, $\hat{\gamma}$, $\hat{\theta}$

2.4 Parameter Estimation

2.4.1 Spatially Varying Coefficients with Sparsity and Homogeneity

The most important aspect of SIRA is our procedure for estimating β. Consider a single covariate j and its corresponding estimate β_j. Because of our sparsity and homogeneity assumptions, β_j represents a configuration of regions of voxels, each taking on distinct values. Given a current region configuration, we would like to modify the regions or their β values to reduce the loss function by as much as possible. We propose the following ways of updating a region configuration, which we call region operations:

1. Expanding - add one voxel that neighbors a region to that region
2. Shrinking - remove one voxel from a region
3. Revaluing - change the β value of a region
4. Merging - combine the voxels of two different regions into a single region, and change the β value of the resulting region
5. Splitting - split one region into two new regions, and change the β values of the two new regions

For any operation that requires changing the β value of a region (revaluing, merging, and shrinking), we require a method to determine what value of β for that region will minimize the loss function. This

problem is equivalent to minimizing a sum of quadratic and piecewise linear functions, and a solution to this class of problems is presented in (Liu & Jiang, 2019). We use the methods presented in this paper to quickly compute optimal β values.

For the merging operation, we restrict the potential merges to regions whose corresponding β values are adjacent to each other when ranked in order, rather than allowing any pair of regions to merge together. For the splitting operation, we only consider splitting one region into two equally sized (or as close as possible) regions. We first compute the gradient of the loss function with respect to the β value of each voxel in the region (a scalar quantity for each voxel), and rank those gradients in order. The voxels ranked in the top half are assigned to the first new region, and the voxels in the bottom half are assigned to the second, with the β values of the resulting regions then optimized via a coordinate descent algorithm. We note that the merging and splitting operations do not guarantee spatial contiguity of the resulting regions; however, in practice, a proposed merge or split yielding spatially discontiguous regions will rarely be accepted as the optimal operation, since the neighboring voxel penalty in the loss function strongly disfavors region configurations with disconnected components. These restrictions on merging and splitting are placed in order to more efficiently identify which operation reduces the loss function the most.

Because we have imposed sparsity and homogeneity, the number of potential region operations for any given region configuration is quite small, and depends on the total number of voxels in regions rather than the total number of voxels in the entire image. Consequently, as long as we have a fast way to evaluate the loss function for any potential region configuration update (section 2.6), we can quickly compute the loss function decrease for every possible region operation and determine which is the best. Then, we accept the operation that decreases the loss function by the most and the resulting region configuration becomes our new estimate for β_j. This region updating algorithm is detailed in Algorithm 2.

We repeat this process of listing out all possible region operations, computing the loss decrease, and accepting the best operation until we reach a point where no region operation further decreases the loss function anymore. At this point, we terminate Algorithm 2 for the current covariate, j, and apply Algorithm 2 to the next covariate, $j+1$. When we have finished applying the β region finding algorithm to all p_1 covariates of interest, we move on to updating $\hat{\gamma}$ and $\hat{\theta}$, then repeat.

Algorithm 2: Region Finding Algorithm for $\hat{\beta}_j$

Input: Current estimates $\hat{\beta}_j$, $\hat{\gamma}$, $\hat{\theta}$; current loss value M
Output: Updated estimate $\hat{\beta}_j$
Set `updated` $\leftarrow$ `true`;
while `updated` **do**
  Set `updated` $\leftarrow$ `false`;
  Enumerate all candidate region operations (expand, shrink, revalue, merge, split) across all regions in $\hat{\beta}_j$ (see Algorithm 3 for more details);
  Let o^* denote the candidate operation yielding the smallest loss M^*;
  if $M^* < M$ **then**
    Execute o^*, set $M \leftarrow M^*$, and sort regions by $\hat{\beta}$ value;
    Set `updated` $\leftarrow$ `true`;
  end
end
return $\hat{\beta}_j$

2.4.2 Other Parameters

Recall that our image-on-scalar regression model is $Y = X\beta + Z\gamma + \theta\psi^* + \epsilon$. Let $\dot{Y}$ denote the quantity $Y - X\beta - \theta\psi^*$, so $\dot{Y} = Z\gamma + \epsilon$. We treat this as a standard linear regression problem, and obtain the least squares estimator

$$\begin{aligned}\hat{\gamma} &= (Z^T Z)^{-1} Z^T \hat{\dot{Y}} \\ &= (Z^T Z)^{-1} Z^T Y - (Z^T Z)^{-1} Z^T X\hat{\beta} - (Z^T Z)^{-1} Z^T \hat{\theta}\psi^*. \end{aligned} \tag{4}$$

We use a similar approach to update our estimate of θ. Let $\tilde{Y}$ denote the quantity $Y - X\beta - Z\gamma$, so $\tilde{Y} = \theta\psi^* + \epsilon$. Also, let Λ denote the $L \times L$ diagonal matrix containing the eigenvalues $\lambda_1, \ldots \lambda_L$ from section 2.1 in descending order. The least squares estimator for θ is

$$\begin{aligned}\hat{\theta} &= \hat{\tilde{Y}}\psi^{*T}(\psi^*\psi^{*T})^{-1} = \hat{\tilde{Y}}\psi^T\sqrt{\Lambda}(\sqrt{\Lambda}\psi\psi^T\sqrt{\Lambda})^{-1} = \hat{\tilde{Y}}\psi^T\sqrt{\Lambda}\Lambda^{-1} = \tilde{Y}\psi^T\Lambda^{-1/2} \\ &= Y\psi^T\Lambda^{-1/2} - X\hat{\beta}\psi^T\Lambda^{-1/2} - Z\hat{\gamma}\psi^T\Lambda^{-1/2} \end{aligned} \tag{5}$$

where we have used the fact that ψ is a semi-orthogonal matrix by construction, i.e., $\psi\psi^T = I_L$.

2.5 Initial Values

Running Algorithm 1 requires initial values for $\hat{\beta}, \hat{\gamma}$, and $\hat{\theta}$. We can begin by setting $\hat{\gamma}$ and $\hat{\theta}$ to 0 and updating their values once required by the algorithm. To initialize the β estimates, we propose using a smoothed MUA-based approach, as that is one of the only methods that can run in a reasonable amount of time for large-scale neuroimaging data.

Let Q denote the block matrix containing all covariates together, i.e., $Q = [X \quad Z]$, and α as the corresponding parameter estimate, i.e., $\alpha = (\beta^\top, \gamma^\top)^\top$. We compute the MUA solution to the image-on-scalar regression, $\hat{\alpha} = (Q^T Q)^{-1} Q^T Y$, as well as the matrix of t-values corresponding to each voxel and covariate. For the covariate of interest j, we apply a smoothing transformation to the α_j to bring t-values of neighboring voxels closer together. We use these smoothed t-values for the local false discovery rate calculation function in the R package locfdr (Efron et al., 2015) and record which voxels have a local false discovery rate estimate below 0.05. Of these voxels, those with positive t-values form our first initial region, and those with negative t-values form our second initial region. The initial β value assigned to these regions is the average value of α_j over the voxels selected for those regions. In the event that locfdr is unable to select any initial regions, we instead rank the smoothed t-values and use the voxels in the top and bottom 2% as our initial regions.

2.6 Computation and Scalability

The most important feature of SIRA is its scalability to large-scale datasets in terms of both number of individuals and voxels. In practice, n and V can reach into the tens or hundreds of thousands, so it is essential that our algorithm never performs any step with a computational complexity of $O(nV)$ or worse. In this section, we highlight the details that allow for rapid computation of our loss function and allow SIRA to run efficiently even for datasets of this scale.

The first strategy we employ to minimize redundant calculations is the usage of summary statistics that can be pre-computed before running Algorithm 1. For example, looking at Equation 4, there are many quantities in this equation that do not depend on quantities that change throughout the algorithm, meaning they can be pre-computed. Namely, $(Z^T Z)^{-1} Z^T Y$ and $(Z^T Z)^{-1} Z^T X$ depend only on fixed, known quantities, so they are computed once and stored at the beginning. Other examples of useful summary statistics are highlighted near the end of this section.

Now, we focus on the mean squared error portion of the loss function, denoted by $L_M(\hat{\beta}, \hat{\gamma}, \hat{\theta})$. Also, let $Y^* = Y - Z\gamma - \theta\psi^*$, allowing us to rewrite our loss function as

$$L_M(\hat{\beta}, \hat{\gamma}, \hat{\theta}) = \frac{1}{n} \sum_{i=1}^{n} \sum_{s_v \in S} \left[\hat{Y_i^*}(s_v) - \sum_{j=1}^{p_1} X_{ij} \hat{\beta}_j(s_v) \right]^2 . \tag{6}$$

This loss function is computationally slow to evaluate, as it contains a double sum over n and V. However, it is much faster to evaluate the difference in the value of the loss function for the current value $\hat{\beta}$, and a candidate value $\hat{\beta}^*$. Note that in Algorithm 2, we only update one covariate at a time, so $\hat{\beta}$ and $\hat{\beta}^*$ will only differ in one row, which we call j^*. The difference in loss, i.e. $L_M(\hat{\beta}^*, \hat{\gamma}, \hat{\theta}) - L_M(\hat{\beta}, \hat{\gamma}, \hat{\theta})$, can be represented as $\frac{1}{n} \sum_{i=1}^{n} \sum_{s_v \in S} \left(\left[\sum_{j=1}^{p_1} X_{ij} \hat{\beta}_j^*(s_v) \right]^2 - \left[\sum_{j=1}^{p_1} X_{ij} \hat{\beta}_j(s_v) \right]^2 - 2\hat{Y_i^*}(s_v) X_{ij^*} \Delta\hat{\beta}_{j^*}(s_v) \right)$, where $\Delta\hat{\beta}_{j^*}(s_v) = \hat{\beta}_{j^*}^*(s_v) - \hat{\beta}_{j^*}(s_v)$. Note that we do not need to worry about $\Delta\hat{\beta}_{j'}(s_v)$ for any $j' \neq j^*$ because those differences will be zero. The first term in this expression (including the sums over n and V) is the squared Frobenius norm of the matrix $X\hat{\beta}^*$, and the second term is the squared Frobenius norm of the matrix $X\hat{\beta}$. Given the value of $\Delta\hat{\beta}_{j^*}$, the difference between these squared norms is easily computable, and only requires summing over the set of voxels whose values are changing, S', not the entire image $\mathcal{S}$. Let $C = \sum_{i=1}^{n} \sum_{s_v \in S'} \left(\left[\sum_{j=1}^{p_1} X_{ij} \hat{\beta}_j^*(s_v) \right]^2 - \left[\sum_{j=1}^{p_1} X_{ij} \hat{\beta}_j(s_v) \right]^2 \right)$. Then,

$$L_M(\hat{\beta}^*, \hat{\gamma}, \hat{\theta}) - L_M(\hat{\beta}, \hat{\gamma}, \hat{\theta}) = \frac{1}{n} [C - \sum_{s_v \in S'} 2(X^T \hat{Y}^*)_{j^*}(s_v) \Delta\hat{\beta}_{j^*}(s_v)].$$

We can consider the $p_1 \times V$ matrix $X^T \hat{Y}^*$ as a summary statistic that needs to be updated whenever $\hat{\gamma}$ or $\hat{\theta}$ are updated. $X^T \hat{Y}^* = X^T(Y - Z\hat{\gamma} - \hat{\theta}\psi^*) = X^T Y - X^T Z\hat{\gamma} - X^T \hat{\theta}\psi^*$. The first term $X^T Y$ is a fixed $p_1 \times V$ matrix and can be pre-computed. For the second term, we can pre-compute the $p_1 \times p_2$ matrix $X^T Z$ and multiply by $\hat{\gamma}$ when it gets updated. The third term is the product of a $p_1 \times n$ matrix, a $n \times L$ matrix, and a $L \times V$ matrix. If we perform the matrix multiplication from left to right, we never have to store an $n \times V$ matrix. We do need to multiply a $p_1 \times L$ matrix by a $L \times V$ matrix, but this is acceptable because $L \ll n$.

It is also simple to compute the loss difference for the other components of the loss function, i.e., the neighboring voxel and L_1 penalties, meaning the entire process of testing whether a region operation decreases the loss function is very fast. It is entirely feasible to test hundreds or thousands of candidate region operations at each step due to the speed at which we can evaluate this loss function difference.

3 Simulations

3.1 Settings

We performed simulation studies to assess the performance of SIRA in comparison to other methods for solving image-on-scalar regression problems. Since other methods do not focus on scalability to the extent that SIRA does, we chose to run simulations on smaller-scale datasets so that all methods would complete without error.

For all of our simulation settings, we chose a sample size of $n = 100$ individuals and generated a 3D cubic image with $V = 8000$ $(20 \times 20 \times 20)$ voxels. There were $p_1 = 2$ covariates of interest, and $p_2 = 2$ covariates of no interest, including a column of ones for the intercept. For both types of covariates, values were generated from a standard normal $\mathcal{N}(0, 1)$ distribution. For the first covariate of interest, we defined two non-overlapping sphere-shaped regions: one with radius 5 centered at the voxel $(6, 6, 6)$, and one with radius 4 centered at the voxel $(15, 15, 15)$. The value of β_1 was set to be 1 inside these regions and 0 otherwise. For the second covariate of interest, we defined a rectangular prism region spanning the voxels (x, y, z) satisfying $3 \leq x \leq 8, 5 \leq y \leq 10$, and $15 \leq z \leq 17$. The value of β_2 was set to be -1 inside this region and 0 otherwise. For the covariates of no interest, all values of γ were drawn from another standard normal distribution.

For each individual, we also generated their spatial individual effect as a Gaussian process, detailed in Section 2.1. This was performed using the R package BayesGPfit (Kang, 2022), which computed the necessary ψ matrix and its corresponding eigenvalues $\lambda_1 \ldots \lambda_L$. The same approaches have been adopted for other image regression models (Lin et al., 2024). For this simulation, the maximum polynomial degree was set to be 8 - consequently, $L = \binom{11}{3} = 165$. We generated θ, as an $n \times L$ matrix with all entries from a standard normal distribution, and set $\eta = \theta\psi^*$. ϵ was generated as an $n \times V$ matrix with all entries as independent zero-mean normal random variables with variances $9, 16$, and 25 for our three settings, corresponding to easy, medium, and hard difficulty settings for our simulation. Finally, our outcome images Y were generated as $Y = X\beta + Z\gamma + \eta + \epsilon$. We generated data under this process 5 times per simulation setting.

We ran SIRA under 12 settings of tuning parameter combinations, with $\lambda \in \{0.1, 0.2, 0.3, 0.5\}$ and $\mu \in \{0.001, 0.01, 0.1\}$. To select the optimal tuning parameter combination, we computed the Bayesian Information Criterion (BIC) of each result, where the degrees of freedom are empirically set as the number of boundary voxels in the solution. The combination resulting in the minimal average BIC was $\lambda = 0.3, \mu = 0.1$. We compared the performance of SIRA to several other image-on-scalar regression

methods - the values of the accuracy metrics are given in Table 1 and a visualization of the results for the medium setting is given in Figure 1. These methods included mass univariate analysis (MUA) and a modification of MUA, mass univariate analysis after pre-smoothing, implemented in Statistical Parametric Mapping (SPM) (Friston, 2003). These are standard and foundational methods used in neuroimaging research, especially for fMRI data. The next method is the bivariate splines over triangulation (BST) method (Li et al., 2021), which is also designed to identify spatially varying regions of constant value. Next is the image response regression via deep neural networks (IRRNN) model (Zhang et al., 2023), designed for capturing complex spatial patterns and straightforward interpretability. Finally, we also look at the Spatially Varying Coefficient Model for Neuroimaging Data with Jump Discontinuities (SVCM) (Zhu et al., 2014), which focuses on producing piecewise smooth, spatially correlated estimates.

3.2 Results

Method	PPV	Sensitivity	IOU	β MSE in Regions
		Easy Setting: $\sigma^2 = 9$		
SIRA	0.9802 (0.0228)	**0.9741 (0.0097)**	**0.9551 (0.0159)**	0.0839 (0.0144)
MUA	0.9519 (0.0091)	0.4984 (0.0889)	0.4854 (0.0822)	0.5498 (0.0913)
SPM	**0.9979 (0.0022)**	0.3534 (0.0910)	0.3532 (0.0910)	0.6711 (0.0945)
SVCM	0.9873 (0.0029)	0.9177 (0.0287)	0.9070 (0.0276)	0.1111 (0.0274)
BST	0.8809 (0.0174)	0.8130 (0.0355)	0.7323 (0.0299)	0.2196 (0.0299)
IRRNN	0.9338 (0.0196)	0.9591 (0.0131)	0.8979 (0.0181)	**0.0642 (0.0153)**
		Medium Setting: $\sigma^2 = 16$		
SIRA	0.9818 (0.0060)	**0.9509 (0.0091)**	**0.9343 (0.0068)**	**0.0935 (0.0059)**
MUA	0.9381 (0.0167)	0.1239 (0.0587)	0.1227 (0.0576)	0.9312 (0.0505)
SPM	**0.9955 (0.0045)**	0.1950 (0.0711)	0.1949 (0.0712)	0.8310 (0.0749)
SVCM	0.9785 (0.0059)	0.8009 (0.0391)	0.7868 (0.0351)	0.2405 (0.0367)
BST	0.8926 (0.0205)	0.6841 (0.0740)	0.6315 (0.0636)	0.3438 (0.0703)
IRRNN	0.9729 (0.0267)	0.6648 (0.1103)	0.6534 (0.1097)	0.3623 (0.1098)
		Hard Setting: $\sigma^2 = 25$		
SIRA	0.9678 (0.0051)	**0.9125 (0.0099)**	**0.8855 (0.0067)**	**0.1119 (0.0107)**
MUA	0.9564 (0.0610)	0.0193 (0.0244)	0.0192 (0.0241)	1.0048 (0.0031)
SPM	0.9916 (0.0102)	0.1411 (0.0698)	0.1411 (0.0698)	0.8845 (0.0712)
SVCM	0.9726 (0.0112)	0.6352 (0.0680)	0.6231 (0.0600)	0.4186 (0.0643)
BST	0.8964 (0.0201)	0.6305 (0.0554)	0.5868 (0.0448)	0.3941 (0.0528)
IRRNN	**0.9925 (0.0047)**	0.4016 (0.0796)	0.4002 (0.0784)	0.6179 (0.0773)

Table 1: Simulation Method Comparison - values are averaged over 5 replications. PPV: positive predictive value, IOU: intersection over union, MSE: mean squared error in regions. SIRA results are for the $\lambda = 0.3, \mu = 0.1$ setting. Results are presented as: Mean (Standard Deviation).

For each simulation setting and method, we computed the following metrics: positive predictive value

(PPV), sensitivity, intersection over union (IOU), and β MSE in regions. Since there are two covariates of interest, all metrics are averaged over the two covariates. To define these metrics, let $A_j = \{s_v : \hat{\beta}_j(s_v) \neq 0\}$ and $B_j = \{s_v : \beta_j(s_v) \neq 0\}$. PPV is the proportion of voxels predicted to have nonzero value that had a true nonzero value, i.e., PPV $= \frac{1}{p_1}\sum_{j=1}^{p_1}\frac{|A_j \cap B_j|}{|A_j|}$. Sensitivity is the proportion of voxels with a true nonzero value that were predicted to have a nonzero value, i.e., Sensitivity $= \frac{1}{p_1}\sum_{j=1}^{p_1}\frac{|A_j \cap B_j|}{|B_j|}$. IOU is the ratio of the cardinality of the set of voxels that were both predicted to be and truly were nonzero to the cardinality of the set of voxels that were either predicted to be or truly were nonzero, i.e., IOU $= \frac{1}{p_1}\sum_{j=1}^{p_1}\frac{|A_j \cap B_j|}{|A_j \cup B_j|}$. β MSE in regions is the mean squared error of $\hat{\beta}$ over voxels which had a true nonzero value, i.e., β MSE in Regions $= \frac{1}{p_1}\sum_{j=1}^{p_1}\frac{1}{|B_j|}\sum_{s_v \in B_j}(\hat{\beta}_j(s_v) - \beta_j(s_v))^2$.

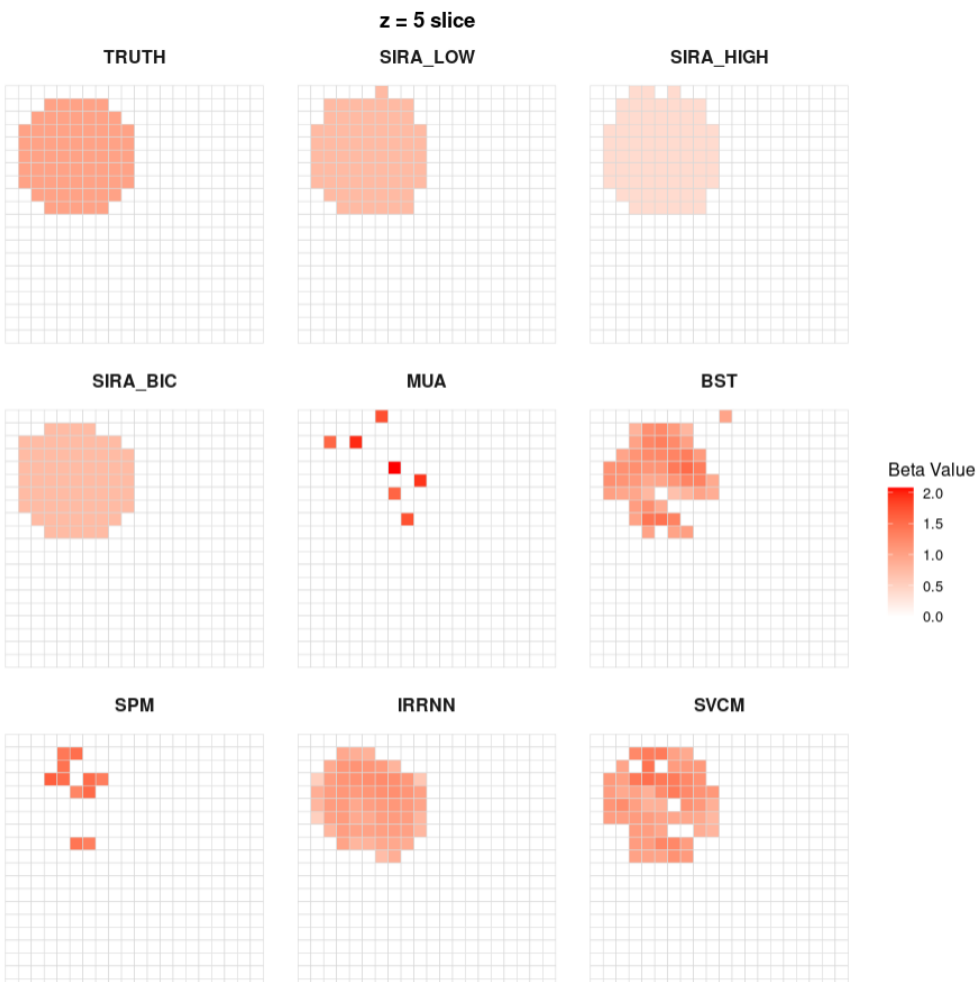

Figure 1: Visualization of the results under the medium error variance setting. Results shown are for the $z = 5$ slice of the coefficient estimate $\hat{\beta}_1$. SIRA_LOW is SIRA with $\lambda = 0.2, \mu = 0.01$. SIRA_HIGH is SIRA with $\lambda = 0.5, \mu = 0.1$. SIRA_BIC is the SIRA solution with the smallest BIC. Here, SIRA_BIC is SIRA with $\lambda = 0.3, \mu = 0.1$.

In every simulation setting, SIRA has by far the greatest sensitivity out of all methods. Although the

PPV is slightly lower than that of SPM or SVCM in certain settings, that is entirely made up for by the much higher sensitivity, which altogether leads to SIRA having the largest IOU value in every setting. Methods like IRRNN and SVCM are able to achieve relatively good values of IOU and β MSE in regions for the easy setting, but as σ^2 increases, the sensitivity of these methods decreases drastically, while SIRA is still able to achieve a sensitivity of over 0.9. For some methods such as MUA or SPM, the sensitivity drops to extremely low values, indicating a lack of power to detect any signal. Since SIRA is able to more accurately identify these nonzero regions, it also nearly uniformly dominates the β MSE in regions column. When considering all metrics simultaneously, SIRA substantially outperforms the other five methods, especially as the error variance increases and other methods begin struggling to identify the signal.

Figure 2: Method running time for various settings, given in seconds. A gray box with no number indicates that the method crashed under that particular setting.

Furthermore, we compared the running time of the six methods while varying the sample size and image

size. All experiments were conducted on a laptop with an AMD Ryzen 5 5600H processor and 8 GB of RAM, running Windows 10. SIRA and MUA were run in RStudio 2025.09.2+418 with R version 4.5.1. BST, SPM, and IRRNN were run using Python 3.9 in PyCharm 2023.2.5. SVCM was run in MATLAB R2023b. The time required for running the methods under each setting is illustrated in Figure 2. SIRA and MUA are the only methods able to produce solutions under all settings, and both achieve substantially faster running times than the other four methods. However, because MUA disregards spatial information, SIRA is the sole method appropriate for real-world high-dimensional imaging data. We note that the computational limitations of SPM are not necessarily fundamental: by accumulating summary statistics of the form X^TX and X^TY across batches of subjects, a scalable implementation of SPM's underlying linear model is achievable. This batching strategy is described for the linear model in (Maullin-Sapey & Nichols, 2021), where it serves as the foundation for their scalable "Big" Linear Mixed Models (BLMM) toolbox, with the underlying product form methodology developed in (Maullin-Sapey & Nichols, 2022).

4 Analysis of UK Biobank Data

The brain imaging data collected by UK Biobank represents one of the most comprehensive and valuable resources for studying the associations between brain structure and individual covariates. One UK Biobank application of interest focuses on task fMRI data obtained during an emotion-related facial recognition paradigm (Miller et al., 2016), based on a study of the human amygdala's response to fearful facial expressions (Hariri et al., 2002). In the faces condition, participants were presented with an image of a face with a fearful expression at the top of the screen and asked to match it to one of two images at the bottom of the screen. In the shapes condition, participants performed an analogous task using geometric shapes, serving as a visual control.

4.1 Data Pre-processing

To apply SIRA, we use the contrast between the faces condition and the shapes condition as the outcome variable. In other words, for each participant the outcome image represents the difference in brain activity between the facial recognition task and the shape-matching control. We treat participant age as our covariate of interest, and include gender, head size, and the interaction between age and gender as covariates of no interest. Continuous variables were centered and scaled to unit variance. The question of interest is to identify regions of the brain where this faces-versus-shapes contrast in activity either increases or decreases with age, after controlling for the covariates of no interest. The entirety of the

fMRI data exists within a rectangular prism of dimension $91 \times 109 \times 91$ voxels, and the sample size is $n = 38,639$. Altogether, this dataset comprised 180 GB when stored in NIFTI format. fMRI data was read and processed in R with the package oro.nifti (Whitcher et al., 2022).

For each voxel within the $91 \times 109 \times 91$ prism, we compute the proportion of individuals with observed data in that voxel. We define our voxel set $\mathcal{S}$ as the collection of voxels that are observed in at least 50% of individuals, and ignore the remaining voxels. This leaves us with a total of $V = 121,865$ voxels for the analysis. We used the Harvard-Oxford cortical and subcortical structural atlases (Desikan et al., 2006) to determine which voxels constituted known regions of the brain. Note that these brain regions are not used during the running of SIRA and are only used to help analyze results after completion of the algorithm.

We then pre-compute various quantities of interest that will facilitate the running of Algorithm 1, including various summary statistics, a list of pairs of neighboring voxels, a sparse adjacency matrix, and the MUA solution along with its corresponding matrix of t-values. Due to the massive size of the outcome variable, imaging data was read in 50 batches, each consisting of images for 773 individuals on average, and summary statistics were cumulatively tallied as each batch was read in. Altogether, this pre-processing procedure was completed in approximately 2 hours.

We ran SIRA with 15 different tuning parameter combinations, intended to cover a wide range of solutions. λ, the tuning parameter for the neighboring voxel penalty, took values from $\{0.001, 0.0025, 0.005, 0.01, 0.025\}$, and μ, the tuning parameter for the global L_1 penalty, took values from $\{0.0005, 0.001, 0.005\}$. For the modified squared exponential kernel (section 2.1), a was set to 0.01, b was set to 100, and h was set to 15, corresponding to $L = \binom{18}{3} = 816$ basis functions. All 15 settings were run in parallel in R on the Great Lakes High Performance Computing Cluster. Jobs were executed on the Intel Xeon Gold 6154 CPUs, using 1 CPU core and 15 GB of memory per job. For the fastest setting, $\lambda = 0.025, \mu = 0.005$, SIRA converged in 1 hour, and for the slowest setting, $\lambda = 0.001, \mu = 0.0005$, SIRA converged in 9.5 hours. The majority of other settings took between 2 to 4 hours to converge.

4.2 Results

We present results from 3 different settings for SIRA as well as the MUA results. These three settings are $(\lambda, \mu) = (0.001, 0.0005), (0.005, 0.001)$, and $(0.025, 0.005)$, corresponding to weak, intermediate, and strong penalty settings, respectively. Specifically for MUA, we applied a false discovery rate correction using the p.adjust R function with the Benjamini-Hochberg method (Benjamini & Hochberg, 1995) to determine which voxels were selected. A visualization of the resulting solutions is displayed in Figure 3.

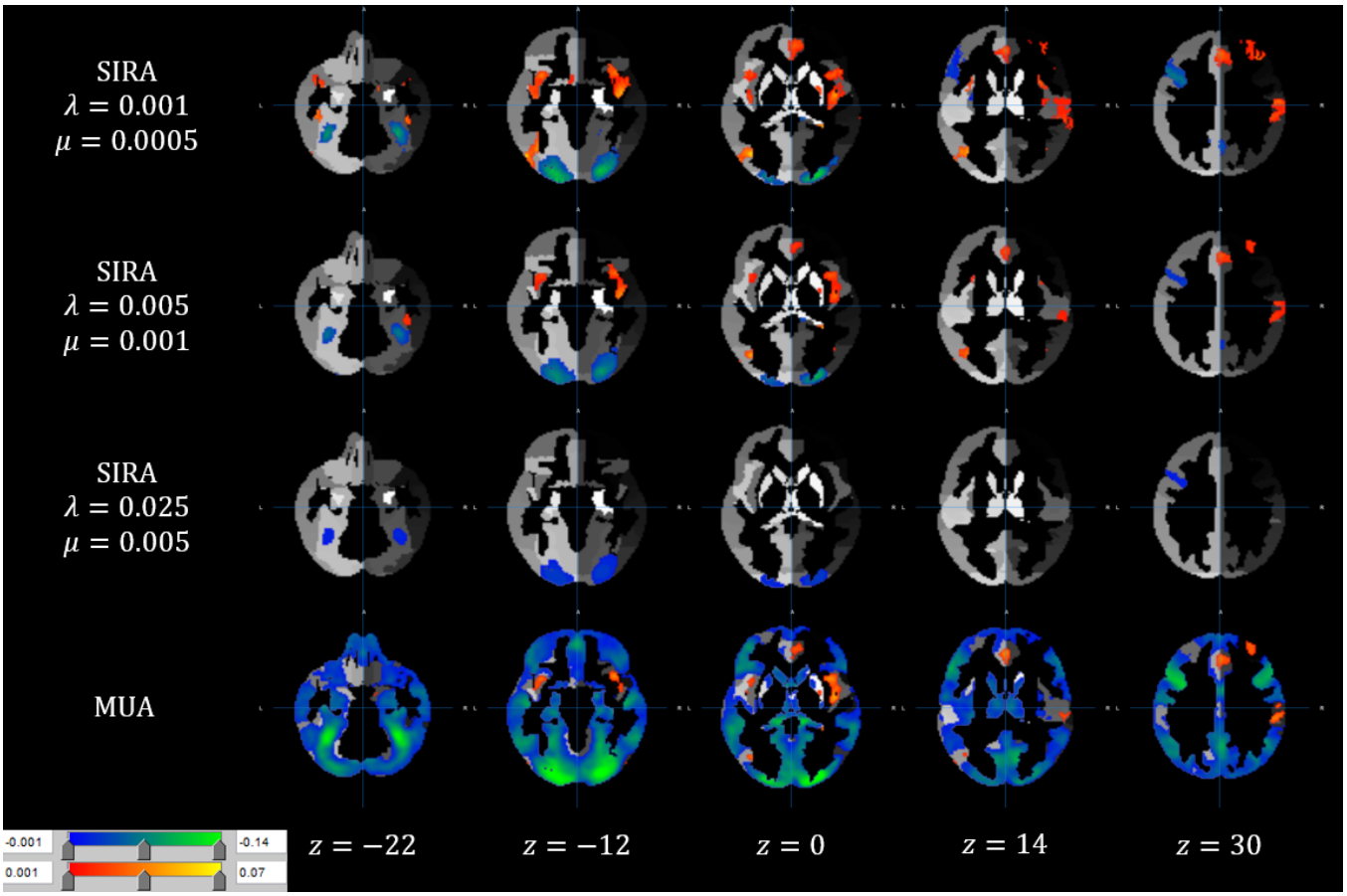

Figure 3: Visualization of the $\hat{\beta}_{\text{age}}$ solutions estimated by SIRA and MUA. Each row represents a different setting/method, each column represents a different axial slice of the brain. Negative values of $\hat{\beta}_{\text{age}}$ are colored on a scale from blue to green (-0.001 to -0.14), and positive values of $\hat{\beta}_{\text{age}}$ are colored on a scale from red to yellow (0.001 to 0.07). Unselected voxels are in grayscale and are colored based on the brain regions defined in the Harvard-Oxford cortical and subcortical structural atlases.

Table 2 displays the numbers of regions and voxels selected for all tuning parameter combinations, and Table 3 highlights the brain regions with the strongest age effect, positive or negative, for the setting where $\lambda = 0.005$ and $\mu = 0.001$.

In Figure 3, we can see a striking difference between MUA and SIRA: MUA selects the vast majority of voxels as significant whereas SIRA is far more selective under the given tuning parameter settings. Several visual patterns that emerge within the MUA solutions are visible within the SIRA solutions, the most notable of these being the two regions in the posterior brain on the $z = -12$ axial slice. These two regions appear consistently in all the SIRA solutions, and we can now visualize the effect of the tuning parameters on these regions. In the $\lambda = 0.001, \mu = 0.0005$ setting, there is a smooth transition from blue to green as one nears the center of these regions, indicating a stronger effect near the centers. Note that in this setting, these two regions are composed of many other smaller regions of constant value,

Method	Lambda (λ)	Mu (μ)	Regions	Voxels
SIRA	0.001	0.0005	92	12,973
		0.001	96	11,237
		0.005	84	7,319
	0.0025	0.0005	95	10,062
		0.001	72	9,571
		0.005	59	6,307
	0.005	0.0005	48	7,663
		0.001	53	7,081
		0.005	40	5,761
	0.01	0.0005	19	6,179
		0.001	15	6,001
		0.005	20	5,516
	0.025	0.0005	2	2,977
		0.001	3	3,032
		0.005	4	2,937
MUA	NA	NA	NA	103,622

Table 2: Number of Regions and Voxels Selected for Each Tuning Parameter Setting

creating a smoothness effect. However, in the $\lambda = 0.025, \mu = 0.005$ setting, the two regions lose all smoothness and become areas of complete constant value. Also, the effect size has been attenuated, as indicated by the darker blue color. This is the effect of strengthening the neighboring voxel and L_1 penalties, respectively. Visual analysis of other axial slices reveals similar results - increasing the tuning parameters creates solutions with less smoothness and fewer selected voxels. This effect is also seen in Table 2, which lists the number of regions and voxels selected by the various settings. The general trend is that increasing either λ or μ tends to result in a solution with fewer regions and voxels selected, as one would intuitively expect. On the other hand, the MUA solution selected approximately 85% of voxels as significant, which is in sharp contrast to the behavior of MUA in our simulation settings. This can be attributed to a much larger sample size in the real data application ($n = 38,639$ vs. $n = 100$), leading to greatly increased power.

In Table 3, we analyze the SIRA solution when $\lambda = 0.005$ and $\mu = 0.001$ against known regions of the brain, and identify the top 5 brain regions ranked by the absolute value of their mean effect size. For example, the first region, the left occipital pole, contains a total of 1,977 voxels, and 34.3% (679) of those were nonzero in $\hat{\beta}_{\text{age}}$. The average value of $\hat{\beta}_{\text{age}}$ across those 1,977 voxels was -0.0183. The interpretation of this value is that a 1 standard deviation increase in age is associated with an average brain signal intensity decrease of 0.0183 standard deviations per voxel within the left occipital pole. The final column lists the mean effect size when restricted to voxels within nonzero regions, i.e., the mean value in this column ignores zeroes, but the previous column includes zeroes in the average.

The top 4 selected regions - left occipital pole, right occipital fusiform gyrus, right occipital pole, and left occipital fusiform gyrus, are indeed the structures that overlap the two prominent regions visible in the $z = -12$ axial slice of Figure 3. The occipital pole is the posteriormost part of the occipital lobe, which is well-known to be responsible for visual processing in the brain. In particular, it is associated with object and face recognition, visuospatial processing, and memory formation (Rehman & Al Khalili, 2023). Additionally, the fusiform gyrus is also known as a key structure for high-level visual processing, including face perception and object recognition (Weiner & Zilles, 2016), and older adults have shown reduced activation in these occipital regions, which is consistent with the negative $\hat{\beta}_{\text{age}}$ values. This aligns exactly with the facial recognition task the individuals were performing as the imaging data was collected, strongly validating the results of SIRA.

5 Discussion

We present SIRA, a novel algorithm for solving image-on-scalar regression problems that is designed for scalability to large-scale datasets in terms of both sample size and image size. By imposing sparsity and homogeneity on the spatially varying coefficient function β, SIRA vastly reduces the number of parameters required to be estimated, enabling it to handle datasets that are far beyond the reach of existing methods. Repeated usage of pre-computed summary statistics and strategies for rapidly evaluating differences in the value of the loss function are essential to SIRA's computational efficiency.

Our simulations provide strong evidence that SIRA outperforms the existing image-on-scalar regression methods MUA, SPM, BST, IRRNN, and SVCM across all accuracy metrics in small-scale settings where all methods are able to run, and further demonstrate that SIRA is the only spatially-aware method capable of scaling to high-dimensional imaging data. Our application to UK Biobank task fMRI data with 38,639 subjects and 121,865 voxels demonstrates that SIRA can produce solutions on real-world

Brain Region Name	**Region Size**	**% Selected**	**Mean $\hat{\beta}_{\text{age}}$**	**Mean $\hat{\beta}_{\text{age}}$ in Regions**
Left occipital pole	1,977	34.3	-0.0183	-0.0533
Right occipital fusiform gyrus	1,450	35.9	-0.0168	-0.0468
Right occipital pole	2,189	28.7	-0.0168	-0.0585
Left occipital fusiform gyrus	1,399	32.7	-0.0146	-0.0446
Right planum polare	338	67.2	0.0144	0.0214

Table 3: Analysis of the brain regions selected by SIRA when $\lambda = 0.005, \mu = 0.001$. The top 5 regions are presented, ranked by absolute value of mean value of $\hat{\beta}_{\text{age}}$ across the brain region. Region size represents the number of voxels in the entire region, % selected represents the percentage of those voxels that are nonzero in $\hat{\beta}_{\text{age}}$.

large-scale datasets within a reasonable time frame. These solutions are easily interpretable and can be combined with brain structural atlases for identification of specific brain regions of interest. In particular, our analysis of participants performing a facial recognition task correctly identified several regions with known roles in visual processing and face perception, validating the utility of SIRA in practice.

It is worth contextualizing the computational comparisons presented in this work. The scalability limitations of SPM demonstrated here are not insurmountable in principle. Through pre-computation of sufficient statistics, parameter estimation for the linear model and linear mixed effects model can be made scalable to large neuroimaging datasets without ever loading the full data into memory simultaneously. SIRA employs a similar batching and summary statistics strategy. However, a fundamental distinction is in the nature of the solution produced. At the scale of the UK Biobank data, MUA selected approximately 85% of voxels as significant, a direct consequence of the greatly increased power at large sample sizes, where high sensitivity is essentially guaranteed for any reasonable method. Since SPM applies Gaussian smoothing prior to a similar voxel-wise analysis, a scalable implementation of SPM would be expected to produce similarly diffuse results at this scale. In this large sample size setting, specificity becomes the more critical metric: the ability to identify a parsimonious, interpretable set of regions rather than flagging the majority of the brain as significant. This is the challenge that SIRA addresses through its assumptions of homogeneity and sparsity on β.

There are two notable limitations of SIRA worth mentioning. First, its running time is strongly tied to the validity of the sparsity assumption on β. In settings where the true β is nonzero over large portions of the brain, the algorithm slows considerably, as the pool of candidate region operations grows with the number of voxels in existing regions. Second, selecting suitable tuning parameters for SIRA requires care. We recommend pre-selecting a range of (λ, μ) combinations, running SIRA in parallel across all combinations, and using a model selection criterion such as BIC, with degrees of freedom empirically set as the number of boundary voxels in the solution, to identify an optimal result. In our simulations, this strategy consistently identified solutions with strong performance in terms of PPV, sensitivity, and β MSE in regions.

Data and Code Availability

The neuroimaging data used in the real data application were obtained from the UK Biobank (www.ukbiobank.ac.uk) under Application Number 34077. UK Biobank data are available to approved researchers upon application at ukbiobank.ac.uk/enable-your-research/apply-for-data. The authors are not permitted to share these data directly. The simulation study used entirely synthetic data generated by

the code described below; no external datasets were required. The SIRA methodology is implemented in the open-source R package SIRA, available at https://github.com/ralphjia/SIRA. The package includes full documentation, a worked vignette, and the code used to generate all simulation results reported in this paper. All analyses were conducted in R version 4.5.1 (R Core Team, 2025) using SIRA version 0.1.0.

Author Contributions

Ralph Jiang: Methodology (supporting), Software (lead), Formal analysis, Validation, Writing – original draft, Writing – review & editing.

Thomas Nichols: Resources (UK Biobank data access), Writing – review & editing.

Hui Jiang: Conceptualization (lead), Methodology (lead), Supervision, Writing – review & editing.

Jian Kang: Conceptualization (lead), Methodology (lead), Supervision, Funding acquisition, Writing – review & editing.

Declaration of Competing Interests

The authors declare no conflicts of interest.

Ethics Statement

This research was conducted using data from UK Biobank, a large-scale biomedical database and research resource containing genetic, lifestyle, and health information from half a million UK participants. UK Biobank has approval from the North West Multi-centre Research Ethics Committee (REC reference 11/NW/0382). All participants provided written informed consent prior to data collection. This research was conducted under UK Biobank Application Number 34077. No additional ethical approval was required for this study, as all data were de-identified and accessed in accordance with the UK Biobank's standard terms of access.

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A Detailed Region Operations Algorithm

We provide a detailed description of the candidate region operations enumerated in Algorithm 2.

Algorithm 3: Candidate Region Operations for Region r (Detailed)

```
Input: Region r; current estimates β̂_j, γ̂, θ̂; current best loss M*; current best operation o*
Output: Updated M* and o*
for each voxel s_v neighboring region r but not within any region do
    Compute loss L if s_v is added to region r;
    if L < M* then
        Set M* ← L, o* ← (expand r by s_v);
    end
end
Compute loss L if region r is revalued to its optimal β̂ value β_r via (Liu & Jiang, 2019);
if L < M* then
    Set M* ← L, o* ← (revalue r to β_r);
end
if region r has at least 2 voxels then
    for each voxel s_v' within region r do
        Compute loss L if s_v' is removed from region r;
        if L < M* then
            Set M* ← L, o* ← (shrink r by s_v');
        end
    end
    Partition voxels of r into two groups by gradient ranking and compute loss L for this split,
     optimizing each group's β̂ via coordinate descent;
    if L < M* then
        Set M* ← L, o* ← (split r);
    end
end
if region r + 1 exists then
    Merge r and r + 1; compute optimal β̂ via (Liu & Jiang, 2019), call it β_r*;
    Compute loss L if r and r + 1 are merged and revalued to β_r*;
    if L < M* then
        Set M* ← L, o* ← (merge r and r + 1);
    end
end
return M*, o*
```